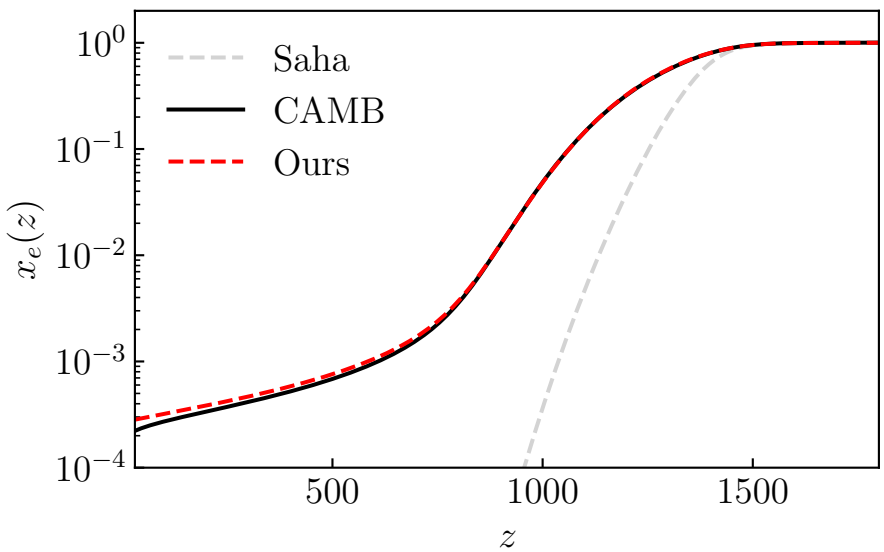
Boltzmann Code - Recombination

The goal of today's lab is to numerically solve for the ionization history around recombination. In particular, we will solve for evolution of the ionization fraction x_e defined as

$$x_e(z) = \frac{n(H^+)}{n(H^0) + n(H^+) \equiv n(H)}.$$

At the end of this lab you should be able to recreate the following plot:



We will make several approximations. The first is that we will ignore helium recombination and the second is that we will assume the baryon temperature is fixed to the photon temperature $T_b(z) = T_\gamma(z)$. For more background, see `lecture_notes/section02.pdf`.

Assignment: In class we discussed the equilibrium treatment of recombination. This results in the **Saha equation**

$$\boxed{\frac{(x_e^S)^2}{1-x_e^S} = \frac{1}{n(H)}\left(\frac{m_e k_B T}{2\pi\hbar^2}\right)^{3/2} e^{-E_I/k_B T},} \tag{1}$$

where x_e^S denotes the ionization fraction in equilibrium and $E_I \approx 13.6$ eV is the binding energy of atomic hydrogen. Throughout this lab we will need the hydrogen number density $n(H)(z)$. Although we ignore helium recombination, helium still removes baryons from the hydrogen budget, so

$$n(H) = \frac{\rho_H}{m_p} = \frac{(1-Y_P)\rho_b}{m_p},$$

where Y_P is the helium mass fraction.

Part 1: Write a function `saha_xe(T, n_H)` which computes x_e given an input baryon temperature T and number density of (neutral and ionized) hydrogen $n(H)$.

As discussed in `lecture_notes/section02.pdf`, things fall out of equilibrium very quickly as soon as the universe becomes slightly neutral ($x_e \approx 0.99$). To properly model this non-equilibrium process we need to solve the Boltzmann equation which you learned in class has the form for a $1+2 \leftrightarrow 3+4$ reaction

$$\frac{1}{a^3}\frac{d(n_1 a^3)}{dt} = -\alpha n_1 n_2 + \beta n_3 n_4.$$

However we can't directly apply this to the reaction $e + H^+ \leftrightarrow H^0_{1s} + \gamma$. This is because recombination directly to the $1s$ state of neutral hydrogen releases a photon of energy E_I which quickly ionizes a nearby neutral hydrogen atom. To avoid this, hydrogen must first recombine to an excited state and then drop to the ground state. For example through

a. **two photon emission**[1] $H(2s) \leftrightarrow H(1s) + \gamma + \gamma$, or

b. **Lyα emission** $H(2p) \leftrightarrow H(1s) + \gamma(\text{Ly}\alpha)$.

Two photon emission channel happens at a rate of (Labzowsky et al 2005)

$$\Lambda_{2s,1s} = 8.2206\,\text{s}^{-1}.$$

The Lyα emission channel, though naively very fast, is suppressed because Lyα photons can get resonantly absorbed by nearby $H(1s)$ atom. However, if the emitted Lyα photon can avoid being absorbed for long enough, it can be redshifted out of the resonance line completely. The rate with which emitted Lyα photons can escape by being redshifted out of the resonance is given by[2]

$$R_{\text{Ly}\alpha} = \frac{8\pi H}{3n(H^0_{1s})\lambda^3_{\text{Ly}\alpha}}.$$

So, the probability that you end up in the $1s$ state given that you start out in a $n = 2$ $(2s, 2p)$ state can then be encapsulated in a factor

$$C \equiv \frac{\frac{3}{4}R_{\text{Ly}\alpha} + \frac{1}{4}\Lambda_{2s,1s}}{\beta_{\text{B}} + \frac{3}{4}R_{\text{Ly}\alpha} + \frac{1}{4}\Lambda_{2s,1s}} \sim \frac{\text{Transition rate from } n = 2 \text{ to } 1s}{\text{Transition rate from } n = 2}, \tag{2}$$

where β_B is the effective photoionization rate. This is called the **Peebles C-factor**. The factor of 3/4 and 1/4 come from $2p$ being a triplet state and $2s$ being a singlet state. After a long calcuation, you will find that the Boltzmann equation we need to solve for x_e is

$$\boxed{\frac{dx_e}{dt} = -C\left(n(H)x_e^2\alpha_B - 4(1-x_e)\beta_B e^{-E_{21}/T_b}\right)} \tag{3}$$

[1] $2s$ and $1s$ state both have zero angular momentum $\ell = 0$ while photons carry angular momentum so to decay from the $2s$ state to the $1s$ state you need to emit at least 2 photons. See your favorite quantum mechanics textbook or these lecture notes.

[2] Note that if we didn't care about Lyα photons being redshifted out of the resonance then Lyα emission would happen much faster than the very slow process of 2γ emission to get from $2s \to 1s$. The bottleneck of resonance escape however slows down the rate of Lyα emission to something comparable to $\Lambda_{1s,2s}$.

where $E_{21} = E(n=2) - E(n=1) = 3E_I/4$. It is much easier to integrate this in redshift so you should work out what dx_e/dz is.

To wrap this up we still need expressions for the coefficient $\alpha_B(T)$ and the effective photoionization rate $\beta_B(T)$. The coefficient α_B is what is typically referred to as the **case-B recombination coefficient**. In particular, α_B is the rate for recombination into any excited state $n \geq 2$; direct-to-ground recombinations are excluded because their photons immediately re-ionize a neighbor. There is no simple closed-form expression, so we use the fitting function of Pequignot, Petitjean & Boisson (1991),

$$\alpha_B(T) = \frac{a\, t_4^b}{1 + c\, t_4^d}\, \mathrm{m}^3\, \mathrm{s}^{-1}, \qquad t_4 \equiv T/10^4\, \mathrm{K},$$

with $a = 4.309 \times 10^{-19}$, $b = -0.6166$, $c = 0.6703$, $d = 0.5300$. The photoionization rate β_B is then fixed by detailed balance:

$$\beta_B(T) = \frac{1}{4}\left(\frac{m_e k_B T}{2\pi\hbar^2}\right)^{3/2} e^{-E_I/(4k_B T)}\, \alpha_B(T).$$

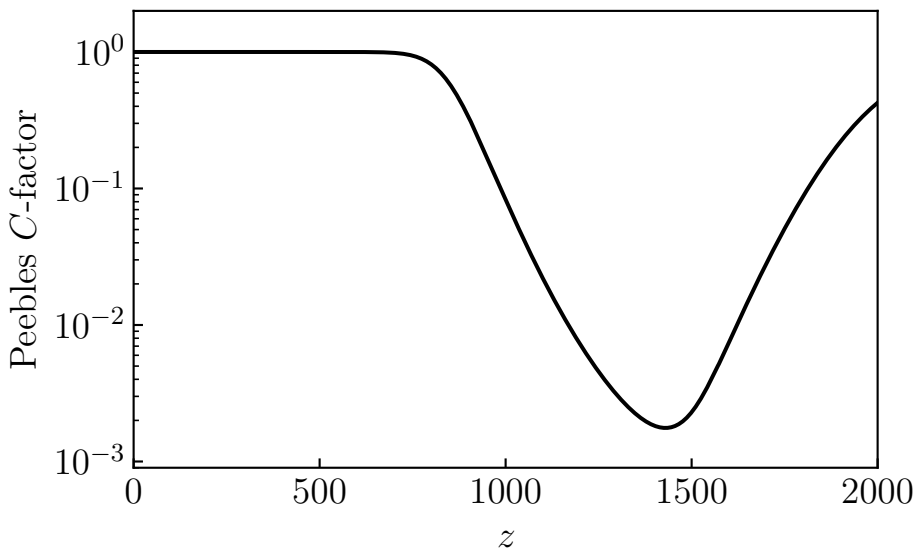
Part 2: Write a function `peebles_rhs(z, x_e, bg)` that returns the right-hand side of (3) rewritten as dx_e/dz.

Once we've implemented `peebles_rhs(z, x_e, bg)`, we are ready to solve for the ionization history.

Part 3: Solve for $x_e(z)$ on a dense redshift grid. The ODE is very stiff at high z, so use Saha equilibrium until $x_e^S(z_\star) \approx 0.99$. Then solve the ODE at $z = z_\star$ with initial condition $x_e(z_\star) = x_e^S(z_\star)$ with `scipy.integrate.solve_ivp`.

Note: I've included in `sol/constants.py` all the constants you should need. You should feel free to just import it and save yourself the trouble of hard coding in the constants.

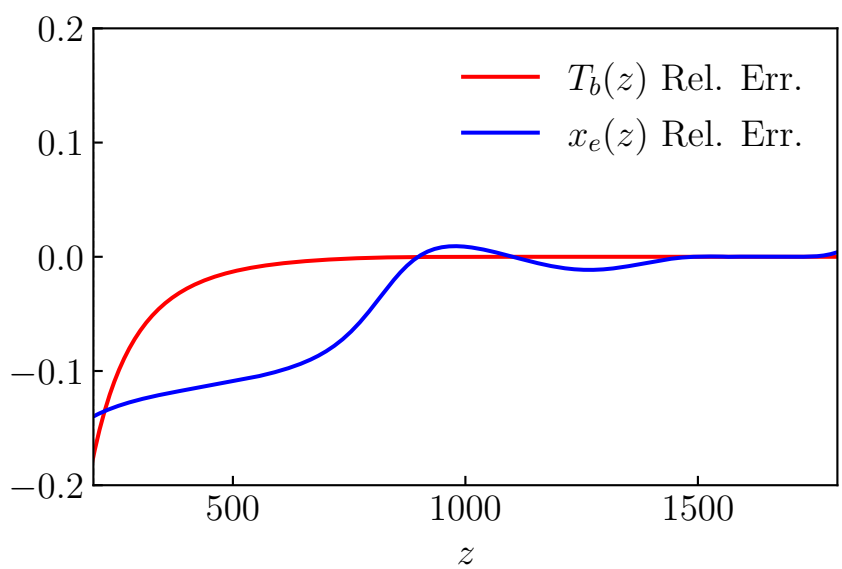
Validation: Lets now validate our code. The first thing that could be interesting to do is plot your computation of the Peebles C-factor and compare this with Fig. 3.16 of Baumann. You should get something like



Your computation of x_e can be compared with **CAMB** (like in the first figure of this handout) by getting **CAMB**'s more accurate ionization history with

```
results.get_background_redshift_evolution(z, ['x_e'], format='dict')
```

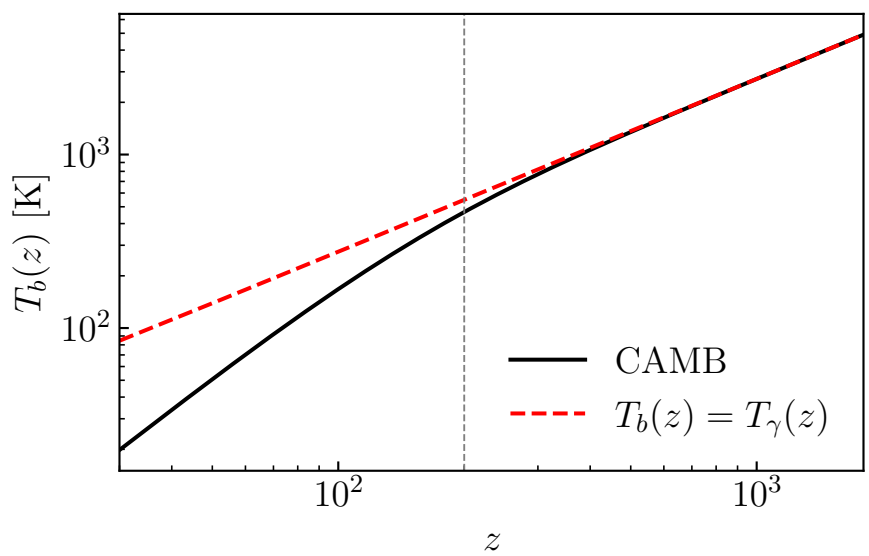
You should find that your computation matches **CAMB** well enough:



A final thing you may be interested in looking at is the temperature evolution of the baryons. We made the approximation in this handout that $T_b(z) = T_\gamma(z)$. However, as discussed in `lecture_notes/section02.pdf`, Compton cooling off the CMB becomes less important than expansion around $z \sim 200$. After expansion takes over, the gas temperature falls as expected from adiabatic expansion, $T_b \propto a^{-2}$. Take a look at

```
results.get_background_redshift_evolution(z, ['T_b'], format='dict')
```

and compare it with the $T_b(z) = T_\gamma(z)$ approximation used in this handout.



If you have extra time this lab, it could be fun to jointly solve for x_e and T_b:

$$\frac{dT_m}{dz} = \frac{2T_m}{1+z} - \frac{\gamma_C(T_\gamma - T_m)}{(1+z)H}, \quad \gamma_C = \frac{8\sigma_T a_r T_\gamma^4}{3m_e c}\frac{x_e}{1+x_e},$$

where γ_C is the Compton coupling rate between the baryons and the CMB photon bath, σ_T is the Thomson cross-section, and $a_r = 4\sigma_B/c$ the radiation constant[3]

[3]Note that once you allow $T_m \neq T_\gamma$, the Peebles ODE (3) also needs a small modification: α_B should be evaluated at T_m (recombination rates are set by the electron kinetic temperature) while β_B and the Boltzmann factor e^{-E_{21}/T_γ} stay at T_γ (photoionization is driven by the photon bath).

References:

- Peebles (1968), *Recombination of the Primeval Plasma*, ApJ 153, 1
  Zel'dovich, Kurt & Sunyaev (1968), *Recombination of Hydrogen in the Hot Model of the Universe*, Sov. Phys. JETP 28, 146 (1969)
  Original papers exploring recombination of neutral hydrogen in the early universe and proposing the three-level atom.

- **Ali-Haïmoud & Hirata (2010)**, *HyRec: A fast and highly accurate primordial hydrogen and helium recombination code*, 1011.3758.
  See Sec. II.A for the Peebles three-level atom derivation in the notation we use here.

- Ma & Bertschinger (1995), *Cosmological Perturbation Theory in the Synchronous and Conformal Newtonian Gauges*, astro-ph/9506072.
  Specifically their section Sec. 5.8 which includes properly tracking Helium recombination.

- Wayne Hu's PhD thesis, *Wandering in the Background: A Cosmic Microwave Background Explorer* (1995), astro-ph/9508126.

For a very pedagogical introduction to recombination physics, I highly recommend the first chapter of Yacine Ali-Haïmoud's PhD thesis, *Cosmological Recombination and Particle Physics* (2011), thesis.caltech.edu/6404.

If you're curious about even more accurate treatments of recombination you should check out `RECFAST` or `HyRec`.